

BullSO 2026 Invitational

Meteorology B Answer Key

Part A: MCQs

1. A
2. D
3. B
4. C
5. B
6. C
7. C
8. C
9. B
10. C
11. A
12. B
13. B
14. A
15. D
16. A
17. A
18. D
19. C
20. C

Part B: Short Answer response

21.

Bands oriented SW to NE suggest steering by the mid-latitude westerlies and alignment with the upper-level jet stream. Precipitation is enhanced along regions of jet streak divergence, which promotes upward motion and cloud development. Describing how the streaks are going towards the direction from SW to NE are also acceptable

22.

Precipitation forecasts are based on model outputs that account for atmospheric moisture, lifting mechanisms, and future cloud development that may not yet be visible on current satellite images.

23.

Estimating the average precipitation within each color band.

For example, green is 0.5–1.0 in → using 0.75 in is an estimate. Students can also use 0.5, 0.6, 0.8, or 1.0 inches. Rounding conversions from inches → meters. Rounding the final multiplication with large areas. Using the calculations should show did:

Green band: $0.01905 \text{ m} \times 2 \times 10^{10} \text{ m}^2 \rightarrow \sim 3.81 \times 10^8 \text{ m}^3$

Yellow band: $0.03175 \text{ m} \times 1 \times 10^{10} \text{ m}^2 \rightarrow \sim 3.18 \times 10^8 \text{ m}^3$

Red band: $0.0635 \text{ m} \times 5 \times 10^9 \text{ m}^2 \rightarrow \sim 3.175 \times 10^8 \text{ m}^3$, All of these values should be added to find the total volume

Total volume $\approx 1.016 \times 10^9 \text{ m}^3$, Allowable range: 1.0×10^9 to $1.05 \times 10^9 \text{ m}^3$

24.

Red and yellow regions (1.0–3.0 in) or areas with poor drainage are more likely to experience localized flooding. Sustained precipitation can saturate soil. Naming the states is also acceptable

25.

Limitation: model uncertainty regarding exact precipitation location, type (rain vs. snow), and intensity, Forecasters combat this by using other satellite models and other sources to corroborate their information

26.

In colorized infrared imagery, yellow to orange (and sometimes red) colors represent the coldest cloud-top temperatures, which correspond to the tallest clouds (often deep convection or strong storm

systems).

These clouds extend higher into the atmosphere, where temperatures are much colder, so they emit less infrared radiation, appearing as brighter warm colors in enhanced IR images.

27.

Linear bands of cold clouds align with warm/cold fronts; surface pressure maps confirm location of air mass boundaries.

28.

In this image, cloud features are distinguished primarily by temperature contrasts, with taller, colder clouds appearing in brighter colors (such as yellow or orange) and lower, warmer clouds or the surface appearing darker. **There is no evidence of sunlight reflection, shadowing, or texture that would typically be visible in visible-light satellite imagery, which requires daylight.** <- Important section to get the point

29.

Areas with warmer surface temperatures and low cloud-top IR readings could cool quickly at night, reaching dew point, producing radiation fog. Students should connect IR cloud-top temperatures, surface cooling, and condensation processes to predict fog.

30.

Linear cloud bands often indicate frontal lifting along a cold or warm front. Could also relate to jet streak divergence aloft, enhancing upward motion and cloud formation.